

Supplementary Materials of RingSketch

1 Proof for Theorem 4.1

Theorem 4.1. *Let $\mathcal{E}$ be the average absolute error (AAE) for estimating the freshness of all items that have appeared within recent $\mathcal{T}$ time units. Under the assumption of no hash collision, we have*

$$\mathbb{E}(\mathcal{E}) \approx \frac{\mathcal{T}_0}{3d}$$

Proof. Under the assumption that there are no hash collision in RingSketch, the estimation error is no more than $\frac{\mathcal{T}_0}{d}$.

Consider a certain item e_i . Suppose RingSketch estimates its pointer position as the midpoint between $\mathcal{A}_j[h_j(e_i)]$ and $\mathcal{A}_{j+1}[h_{j+1}(e_i)]$ where $h_j(e_i), h_{j+1}(e_i) \in [0, m)$. Denote $h_j(e_i)$ as x_j and $h_{j+1}(e_i)$ as x_{j+1} . The probability that the pointer locates between $\mathcal{A}_j[h_j(e_i)]$ and $\mathcal{A}_{j+1}[h_{j+1}(e_i)]$ at e_i 's last arrival is

$$\mathcal{P}_j = \frac{x_j - x_{j+1} + m}{md}$$

The average estimation error of e_i is

$$\overline{\Delta F_i} = \frac{\mathcal{T}_0 \cdot (x_j - x_{j+1} + m)}{4md}$$

We derive $\mathbb{E}(\mathcal{E})$ by integrating over x_j and x_{j+1} as

$$\mathbb{E}(\mathcal{E}) \approx \frac{1}{d} \sum_{j=1}^d \frac{1}{m^2} \int_0^m \int_0^m \mathcal{P}_j \cdot \overline{\Delta F_i} dx_j dx_{j+1} = \int_{-1}^0 \int_0^1 (y - x)^2 \cdot \frac{\mathcal{T}_0}{4d} dy dx = \frac{\mathcal{T}_0}{3d}$$

□

2 Proof for Theorem 4.2

Theorem 4.2. *Consider a certain item e_i that has appeared within recent $\mathcal{T}$ time units. Suppose there are w distinct items in the data stream between the last arrival time of e_i and the present moment. Let $\Delta F_i = |F_i - \hat{F}_i|$ be the estimated error of RingSketch for the freshness of e_i . We have*

$$\mathbb{E}(\Delta F_i) \leq \frac{\mathcal{T}_0}{3d} + (1 - \mathcal{P}) \cdot \frac{5\mathcal{T}_0}{12d} + (1 - \mathcal{P})^d \cdot \frac{\mathcal{T}}{2}$$

where $\mathcal{P} = (1 - \frac{1}{m})^w \approx e^{-\frac{w}{m}}$ is the probability that e_i does not collide with other item in any of the d parts, and $\mathcal{T}_0 = \frac{dm}{V}$ is the time for the pointer to complete one cycle.

Proof. We discuss the estimation error of RingSketch in the following three cases.

Case 1: e_i doesn't collide with other items in all the d parts. In such case, the estimation error is resulted from the difference between the estimated pointer position at e_i 's last arrival time and the true pointer position, which is no more than $\frac{1}{d}$ cycle. According to the conclusion of Theorem 4.1, the average estimation error is $\frac{\mathcal{T}_0}{3d}$.

Case 2: e_i collides with other items in some of the d parts. In such case, the estimation error is also resulted from the estimation difference of the pointer position. Suppose e_i has hash collisions in j parts ($1 \leq j < d$). The estimation difference is no more than $\frac{(j+2)\mathcal{T}_0}{2d}$, whose average is $\frac{(j+2)\mathcal{T}_0}{4d}$.

Case 3: e_i collides with other items in all the d parts. In such case, we cannot accurately infer the number of rotations the pointer has made since e_i 's last arrival. Therefore, the estimation error can be up to $\mathcal{T}$, whose average is $\frac{\mathcal{T}}{2}$.

Now we consider the probability $\mathcal{P}$ that e_i does not collide with other item in any of the d parts. For a certain item e_j , the probability that it does not collide with e_i at $\mathcal{A}_1[h_1(e_i)]$ is $1 - \frac{1}{m}$, where m is the number of counters in $\mathcal{A}_1$. As there are w distinct items between e_i 's last arrival to the present, we have $\mathcal{P} = (1 - \frac{1}{m})^w \approx e^{-\frac{w}{m}}$.

As the d parts of RingSketch are independent from each other, the number of collisions j obeys the binomial distribution $\mathcal{B}(d, \mathcal{P})$. Therefore, the expected error upper bound can be calculated as

$$\begin{aligned}\mathbb{E}(\Delta F_i) &\leq \frac{\mathcal{T}_0}{3d} \cdot \mathcal{P}^d + \sum_{j=1}^{d-1} \binom{d}{j} \mathcal{P}^{d-j} (1-\mathcal{P})^j \cdot \frac{(j+2)\mathcal{T}_0}{4d} + (1-\mathcal{P})^d \cdot \frac{\mathcal{T}}{2} \\ &\leq \frac{\mathcal{T}_0}{3d} + \frac{5(1-\mathcal{P})\mathcal{T}_0}{12d} + (1-\mathcal{P})^d \cdot \left(\frac{\mathcal{T}}{2} - \frac{(d+2)\mathcal{T}_0}{4d} \right) \\ &\leq \frac{\mathcal{T}_0}{3d} + (1-\mathcal{P}) \cdot \frac{5\mathcal{T}_0}{12d} + (1-\mathcal{P})^d \cdot \frac{\mathcal{T}}{2}\end{aligned}$$

□

3 Proof for Lemma 4.1

Lemma 4.1. Consider a data stream following Zipf distribution with N items derived from n distinct items and parameter α . Let e_k be the k_{th} most frequent item in the Zipf distribution, and w_k be the number of distinct item in the data stream since e_k 's last arrival. We have

$$\mathbb{E}(w_k) = \sum_{j=1}^n \frac{j^{-\alpha}}{j^{-\alpha} + k^{-\alpha}} - \frac{1}{2}$$

In particular, let $N_{\mathcal{T}}$ be the number of items that have appeared within recent $\mathcal{T}$ time units.

When $\alpha = 1.0$, we have

$$\mathbb{E}(w_k) \leq \min \left(k \cdot \ln \left(\frac{n}{k} + 1 \right) - \frac{1}{2}, \frac{N_{\mathcal{T}}}{2} \right)$$

When $\alpha = 1.5$, we have

$$\mathbb{E}(w_k) \leq \min \left(2.42k - \frac{1}{2}, \frac{N_{\mathcal{T}}}{2} \right)$$

Proof. Consider the j_{th} most frequent item e_j . In a Zipf distribution, the frequencies of e_j and e_k are $\frac{N}{j^\alpha \zeta(\alpha)}$ and $\frac{N}{k^\alpha \zeta(\alpha)}$ respectively. Therefore, the probability that e_j appears after the last appearance of e_k is

$$p_j = \frac{j^{-\alpha}}{j^{-\alpha} + k^{-\alpha}}$$

.

We can calculate $\mathbb{E}(w_k)$ by summing up p_j as

$$\mathbb{E}(w_k) = \sum_{j=1 \wedge j \neq k}^n p_j = \sum_{j=1}^n \frac{j^{-\alpha}}{j^{-\alpha} + k^{-\alpha}} - \frac{1}{2}$$

Therefore, we have

$$\mathbb{E}(w_k) = \sum_{j=1}^n \frac{1}{1 + \left(\frac{j}{k}\right)^\alpha} - \frac{1}{2} \leq \int_0^{\frac{n}{k}} \frac{k}{1+t^\alpha} dt - \frac{1}{2}$$

When $\alpha = 1$, we have

$$\mathbb{E}(w_k) \leq k \ln \left(\frac{1 + \frac{n}{k}}{1 + 0} \right) - \frac{1}{2} = k \ln \left(\frac{n}{k} + 1 \right) - \frac{1}{2}$$

When $\alpha = 1.5$, we have $\int_0^{\frac{n}{k}} \frac{1}{1+t^\alpha} dt \approx 2.42$, meaning that

$$\mathbb{E}(w_k) \leq 2.42k - \frac{1}{2}$$

Notice that as there are $N_{\mathcal{T}}$ items that have appeared within recent $\mathcal{T}$ time units, the expectation of w_k cannot be larger than $N_{\mathcal{T}}$ on average. Therefore, we also have

$$\mathbb{E}(w_k) \leq \frac{N_{\mathcal{T}}}{2}$$

□

4 Proof for Theorem 4.3

Lemma 4.3. Consider a data stream following Zipf distribution with N items derived from n distinct items and parameter α . Let $\mathcal{E}$ be the average absolute error (AAE) of RingSketch for estimating the freshness of all items that have appeared within recent $\mathcal{T}$ time units. The upper bound of $\mathbb{E}(\mathcal{E})$ satisfies

$$\mathbb{E}(\mathcal{E}) \leq \frac{\mathcal{T}_0}{3d} + \sum_{k=1}^n \frac{k^{-\alpha}}{\zeta(\alpha)} \cdot \left(\frac{5\mathcal{T}_0 w_k}{12md} + \left(\frac{w_k}{m} \right)^d \cdot \frac{\mathcal{T}}{2} \right)$$

where w_k is the number of distinct item in data stream since e_k 's last arrival, which can be substituted by the expectation (or upper bound) in Lemma 4.1 to attain an upper bound.

Proof. For each item in a data stream following Zipf distribution, the probability that it is the k_{th} most frequent item e_k is

$$q_k = \frac{1}{k^\alpha \zeta(\alpha)}$$

According to Theorem 4.2, we can derive the average error upper bound for all items as

$$\begin{aligned} \mathbb{E}(\mathcal{E}) &\leq \frac{\mathcal{T}_0}{3d} + \sum_{k=1}^n q_k \cdot \left(\frac{5\mathcal{T}_0(1-\mathcal{P})}{12d} + \frac{(1-\mathcal{P})^d \cdot \mathcal{T}}{2} \right) \\ &= \frac{\mathcal{T}_0}{3d} + \sum_{k=1}^n \frac{k^{-\alpha}}{\zeta(\alpha)} \cdot \left(\frac{5\mathcal{T}_0}{12d} \left(1 - e^{-\frac{w_k}{m}} \right) + \frac{\mathcal{T}}{2} \left(1 - e^{-\frac{w_k}{m}} \right)^d \right) \\ &\leq \frac{\mathcal{T}_0}{3d} + \sum_{k=1}^n \frac{k^{-\alpha}}{\zeta(\alpha)} \cdot \left(\frac{5\mathcal{T}_0 w_k}{12md} + \left(\frac{w_k}{m} \right)^d \cdot \frac{\mathcal{T}}{2} \right) \end{aligned}$$

□